Merge Sort & Merging Techniques

Question: Given 2 sorted arrays, merge them to get a sorted array.

ex: $A = [1, 5, 6, 9]$ # N

$B = [2, 4, 8]$ # M

$O/P = [1, 2, 4, 5, 6, 8, 9]$

Brute Force Approach

- Concatenate the arrays
- Sort merged array

$TC = O((N+M) * \log(N+M))$

Optimised

i

$A = [1, 5, 6, 9]$

$B = [2, 4, 8]$

j

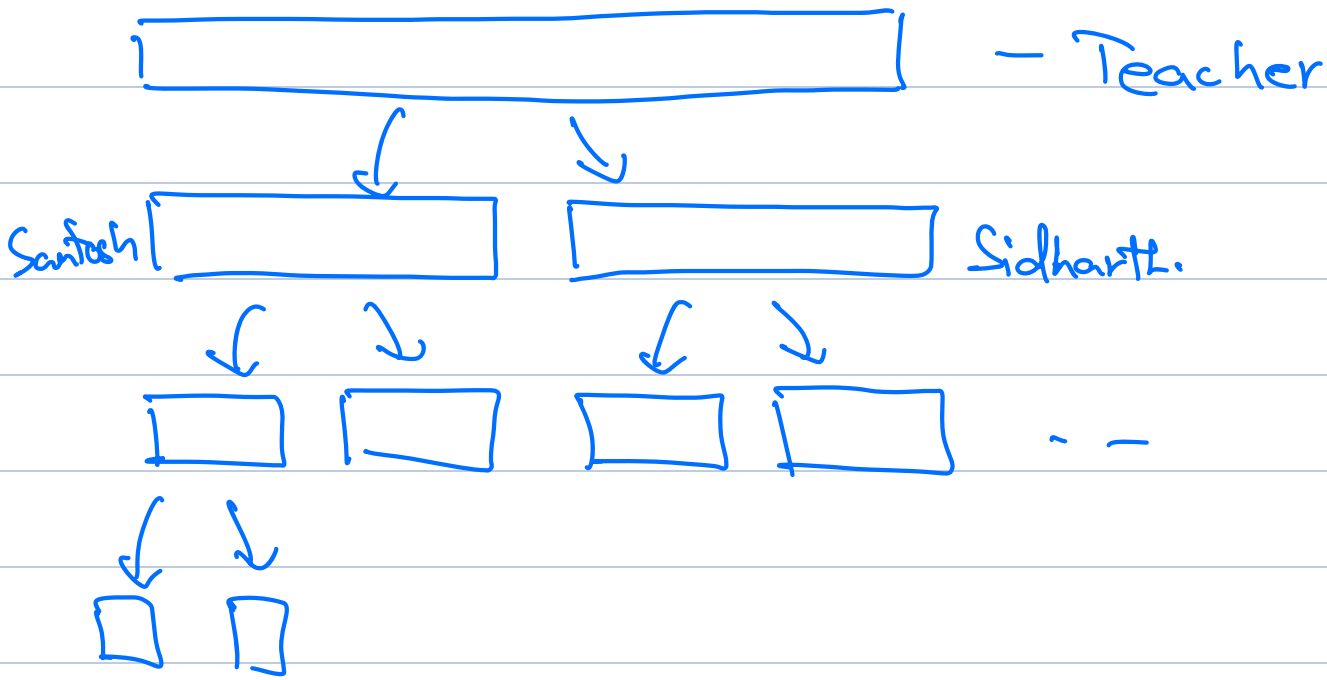
$O/P = [1, 2, 4, 5, 6, 8, 9]$

k

```
def merge(A, B):
    i=0 , j=0
    out = [0]* (len(A) + len(B))
    k=0
    while( i < len(A) and j < len(B)):
        if( A[i] <= B[j]):    # Stable sort.
            out[k] = A[i]
            i+=1
            k+=1
        else
            out[k] = B[j]
            j+=1
            k+=1
    while( i < len(A)):
        out[k] = A[i]
        i+=1
        k+=1
    while( j < len(B)):
        out[k] = B[j]
        j+=1
        k+=1
    return out
```

$$\boxed{\begin{array}{l} TC = O(N+M) \\ SC = O(1) \end{array}}$$

Merge Sort



0	1	2	3	4	5	6	7	8

ex: [3, 10, 6, 8, 15, 2, 12, 18, 17]

[2, 3, 6, 8, 10, 12, 15, 17, 18]

[3, 10, 6, 8, 15] [2, 12, 18, 17]

[3, 6, 8, 10, 15] [2, 12, 17, 18]

[3, 10, 6] [8, 15] [2, 12] [18, 17]

[3, 6, 10] [8, 15] [2, 12] [17, 18]

[3, 10] [6] [8] [15] [2] [12] [18] [17]

[3, 10]

[3] [10]

```
def mergeSort(arr):
    # Base Case
    if(len(arr)==1): return arr.
    # Main logic
    m= len(arr)//2
    A = arr[:m+1]
    B = arr[m+1:]
    A= mergeSort(A)
    B= mergeSort(B)
    out = merge(A,B)
    return out.
```

TC & SC

$N \rightarrow N$

$N/2 \quad N/2 \rightarrow N$

$N/4 \quad N/4 \quad N/4 \quad N/4 \rightarrow N$

- - - - - - - -

1 --- 1 1 --- 1 1 --- 1

$N \rightarrow N/2 \rightarrow N/4 \rightarrow \cdots 1 \Rightarrow \log N \rightarrow$ <u>Levels</u>

$TC = O(N \log N)$

$TC = O(\#\text{ of func}^n \text{ calls} * \text{Time per func}^n \text{ call})$.

	N			→	1
N/2		N/2		→	2
N/4	N/4	N/4	N/4	→	4
---	---	---	---		
1 --- 1	1 --- 1	1 --- 1		→	--

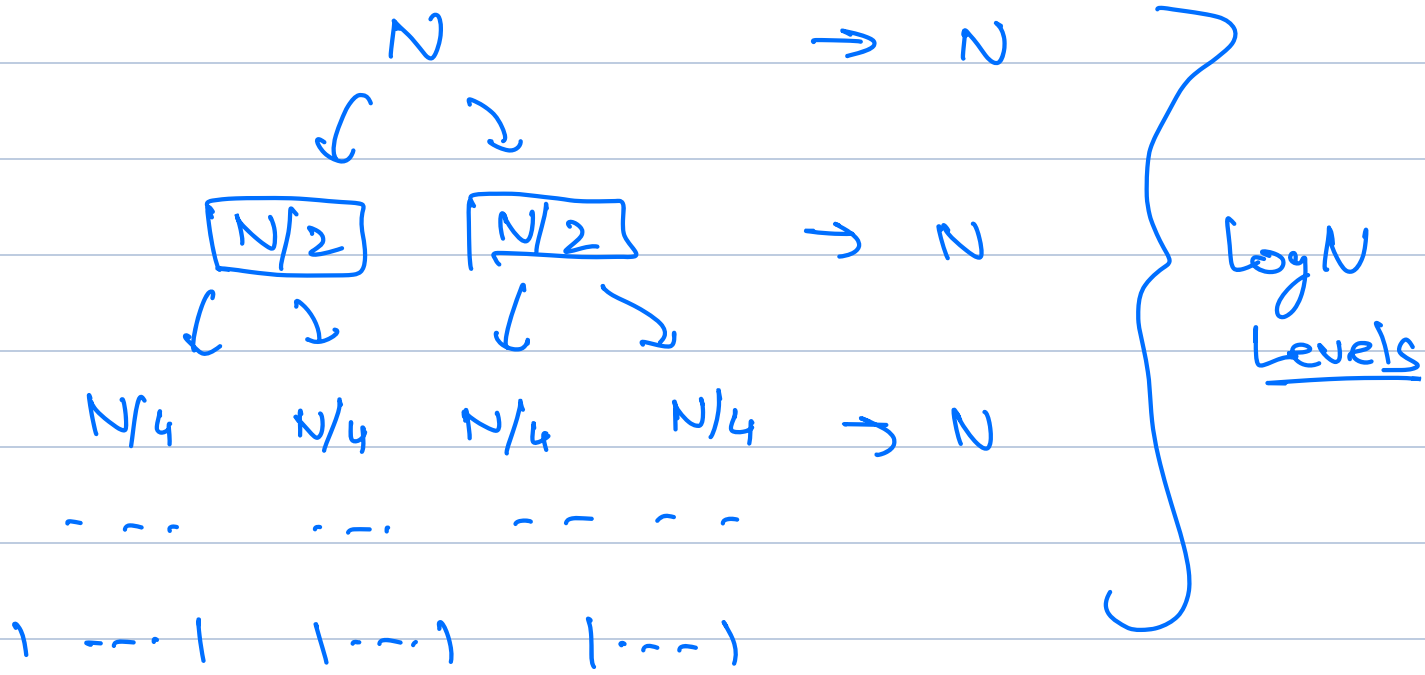
<u>Levels</u>

$1 + 2 + 4 \cdots$

$$\text{Sum} = \frac{1 * (2^{\log N} - 1)}{2 - 1} = N$$

$TC = O(N * \log N)$

Space Complexity



Total Space = $O(N + \log N) = O(N)$

$SC = O(\text{Max depth} + \text{Space per func}^n \text{ call})$.

$= O(\log N + N)$

$SC = O(N)$

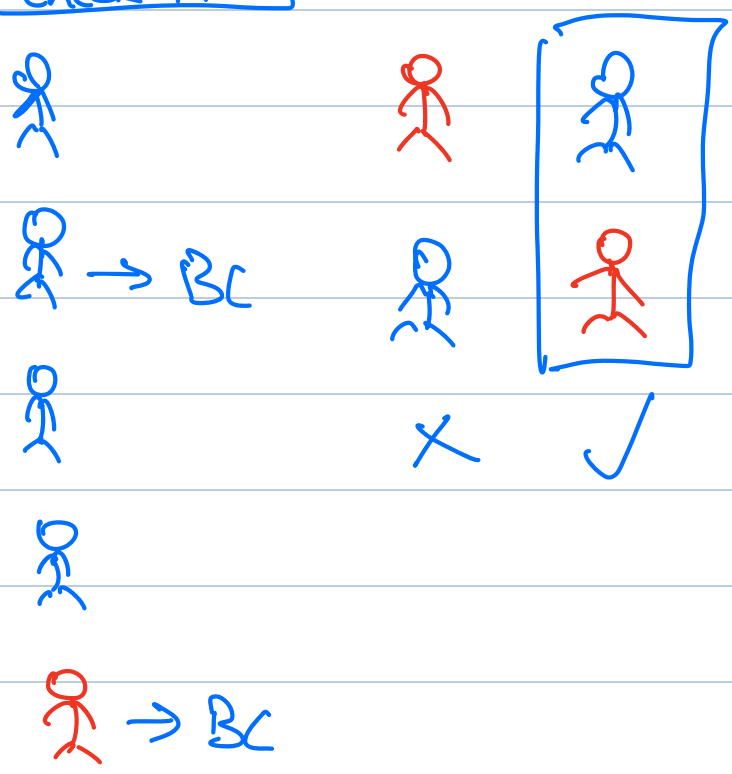
Stable Sort : Respecting the insertion order if the magnitude is the same.

[4 , 3 , 5 , 1 , 2 , 5]

⇓

[1 , 2 , 3 , 4 , 5 , 5] ⇒ Stable Sorting

Check-in



Inplace Sorting : No extra space consumed.

Merge Sort is not inplace sort.

JSON → JavaScript Object Notation (Web-based).

Dict → {"a" : 1 , "b" : 2}

JSON

```
{
    "a" : 1,
    "b" : 2
}
```

```
import json
d = {"a" : 1 , "b" : 2}
json_out = json.dump(d)   # Dict → JSON.
print(json_out["a"])   # 1
d = json.load(json_out)
```

YAML : Config settings for desktop applications.

Dict → {"a" : 1 , "b" : 2}

YAML

```
a : 1
b : 2
```

"a:1 \n b:2" = yaml_text.

```
import yaml
d = yaml.safe_load(yaml_text)
yaml_out = yaml.dump(d)
```

```
print(yaml_out["a"])  # 1.
```

Argparser

```
python3 <file_name>
```

↳ No parameters passed.

```
abc
...
...
config_setting = fromfile
...
...
```

With argparser

```
python3 <file_name> abc.yaml.

python3 <filename> --help.
```

file.py

```
import argparse

parser = argparse.ArgumentParser()
parser.add_argument("a", type=int, help="First number")
parser.add_argument("b", type=int, help="Second number")
args = parser.parse_args().  # Load command
```

line arguments.

```
print("Sum: ", args.a + args.b)
```

Command line

```
python3 file.py --a 1 --b 4
# Sum: 5

python3 file.py # Error from argparse
```

To declare default values

```
- - -

parser.add_argument("a", type=int, default=5,
                    help="First number")
parser.add_argument("b", type=int, default=1,
                    help="Second number")

- - -
```

Command line

```
python3 file.py.
# Sum: 6
```

Passing boolean command line argument

```
import argparse
parser = argparse.ArgumentParser()
parser.add_argument("name", help="Your name")
parser.add_argument("shout", action="store_true",
                    help="Uppercase greeting")
args = parser.parse_args()
if args.shout:
    print("HI", name.upper())
else:
    print("Hi", name.capital())
```

Command line

```
python3 file.py --name="Bhargav" --shout
# HI BHARGAV

python3 file.py --name="Bhargav"
# Hi Bhargav
```